

OCR A-Level Computer Science

Computer Systems (Component 01)

Practice Exam: The characteristics of contemporary processors, input, output and storage devices

Time allowed: 1 hour

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above.
- Answer ALL questions.
- Write your answers in the spaces provided.
- Additional paper may be used if required.

INFORMATION FOR CANDIDATES

- The total mark for this paper is 40.
- The marks for each question are shown in brackets [].
- Quality of extended responses will be assessed in questions marked with an asterisk (*).
- This document consists of 8 pages.

Name:	Centre Number:
Candidate Number:	Date:

Grade Boundaries:

- A*: 36-40 marks
- A: 32-35 marks
- B: 28-31 marks
- C: 24-27 marks
- D: 20-23 marks

Question 1

- a. Explain what is meant by the term 'clock speed' in a processor. [2 marks]

- b. State two factors, other than clock speed, that affect the performance of a processor. [2 marks]

- c. Explain the difference between RISC and CISC processor architectures. [4 marks]

Question 2

- a. Describe the purpose of cache memory in a processor. [2 marks]

- b. Explain the relationship between L1, L2, and L3 cache. [3 marks]

- c. A processor has a clock speed of 3.6 GHz. Calculate how many cycles this processor can execute in one second, showing your working. [2 marks]

Question 3

a. Describe the purpose of each of the following components in the Fetch-Decode-Execute cycle:

i. Program Counter (PC) [2 marks]

ii. Memory Address Register (MAR) [2 marks]

iii. Memory Data Register (MDR) [2 marks]

b. Describe the stages of the Fetch-Decode-Execute cycle. [4 marks]

Question 4

a. Explain what is meant by 'pipelining' in processor architecture. [3 marks]

b. Describe one advantage and one disadvantage of pipelining. [4 marks]

Question 5*

A computer manufacturer is designing a new laptop aimed at graphic designers. The laptop needs to handle complex image processing tasks efficiently.

Discuss the processor characteristics and storage devices that would be most appropriate for this laptop. Justify your choices and explain how they would benefit graphic designers. [8 marks]

End of Questions

Total: [40 marks]